

Adharsh Shokkalingam

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EDUCATION

University of Alberta, Faculty of Computing Science

September 2024 – May 2027

Bachelor of Science; Major in Computing Science Artificial Intelligence, Minor in Mathematics

- **Relevant coursework:** Algorithms, Software Engineering, Computer Architecture, Operating Systems, Machine Learning, Reinforcement Learning, Databases, Calculus I, II, III, IV.

MacEwan University, Faculty of Science

September 2022 – August 2024

Bachelor of Science; Major in Mathematics, Minor in Computing Science

EXPERIENCE

AlbertaSAT

Edmonton, AB

Incoming Software Team Volunteer

May 2026 – Current

- Will be developing flight software for a university Ex-alta3 mission in Rust, targeting bare metal embedded systems with strict reliability constraints
- Interfacing across embedded hardware and software sub teams in an open source satellite codebase to integrate and validate mission critical components

PROJECTS

3D Engine CPU Rasterizer – C, SDL2

June 2025 – Augst 2025

- Engineered a fully interactive **3D engine** in C, leveraging **SDL2** for graphics abstraction and input handling, with backface culling, near plane clipping, dynamic camera transformation, and basic lighting.
- Designed multiple **real-time rendering modes (wireframe, vertex-highlighted, solid-fill, textured)** with keyboard-driven mode switching and **6** degrees of camera freedom.
- Architected the project into **modular components (lighting, geometry, mesh, culling, camera, etc), enabling extensibility and low level control over the render pipeline.**

Sulfur – Events Lottery Application – Java, Firebase

September 2025 – December 2025

- Developed a full stack **android** application with **6** group members using **Java, android studio, Firebase** for managing and **tracking** event/user data.
- Implemented **QR Code** generation and scanning features including the storing and extraction of embedded **geolocation** data to display and update event/users location.
- Utilized **Firebase Cloud Firestore** for structured cloud data storage, **querying**, and **multi-user** data consistency.

RISC-V to WASM Converter - RISC-V (RV32I)

September 2026 – December 2026

- Designed and implemented support for **R-type, I-type, and Branch instructions**, including arithmetic, logic, shifts and **conditional control flow**.
- Built a **signed LEB128 encoder** from scratch to extract immediate operands and convert to **WASM** byte code.

Distributed System Monitor - C++

January 2026 – May 2026

- Built a distributed monitoring system across two programs distpsnotify and remoteagent, coordinating real time data collection over TCP/IP sockets from multiple remote machines simultaneously.
- Designed and implemented a custom client server command protocol (EXECUTE/QUIT, START/STOP delimiters) for structured, ordered data exchange across persistent socket connections.
- Engineered dynamic memory management for an arbitrary number of remote agents using heap allocation, verified zero memory leaks across all code paths using Clang's leaks tool on macOS.

Health Monitor - ESP32-S3, C++, FreeRTOS, LVGL (IP)

May 2026 – Current 2026

- Building a wrist worn health monitor integrating a MAX30102 optical biosensor for real time heart rate and SpO2, a 6 axis IMU for step counting, and a live scrolling ECG display across dual FreeRTOS tasks on an ESP32-S3.

Rust OS – RUST, QEMU

May 2026 – Current 2026

- Building a bare metal x86_64 OS kernel with custom IDT for CPU exceptions and hardware interrupt handling, multilevel paging for virtual memory, and a heap allocator supporting bump, linked list, and fixed size block strategies.

SKILLS & CERTIFICATIONS

Technical Skills: Python, Java, C/C++, RISC-V Assembly (RV32I), Rust, SPSS, HTML, CSS

Developer Tools: Git, VS Code, Firebase, UML, Android Studio, SDL2, SQL, MongoDB, Figma

Certifications: University of Alberta Specialization Reinforcement Learning

Interests: Travel, Hiking, Weightlifting, Mathematics